

BlockHouse Trial Task

Alex Pyo

May 2025

1. **Motivation behind measuring OFI at multiple depth levels of the order book:**

The best-level OFI captures net supply/demand at the inside quotes, but limit orders placed deeper in the book also reflect latent buying or selling pressure that can predict subsequent price moves. Empirical studies show that including OFIs from levels 1 through 10 increases in-sample R^2 from about 71% (using only level 1) to nearly 89%, and out-of-sample performance similarly improves. Thus, multi-level OFI uncovers additional informational content beyond the top of book.

2. **Why the authors use LASSO rather than OLS for estimating cross-impact:**

Cross-impact models include one OFI regressor per asset, so with hundreds of stocks but only tens of observations per rolling window, ordinary least squares is ill-posed and severely overfits when predictors are highly correlated. LASSO adds an ℓ_1 penalty that induces sparsity, automatically selecting the few assets whose order-flow imbalances truly affect the target stock's return, yielding more stable and interpretable cross-impact coefficients.

3. **Why OFI is a better predictor of short-term returns than trade volume:**

Trade volume aggregates all buys and sells but does not distinguish between aggressive and passive orders or capture changes in resting liquidity. OFI, by construction, measures the net change in bid and ask depth (and price shifts) at each update, directly quantifying supply versus demand pressure. This directional imbalance aligns more closely with immediate price moves and explains a much larger fraction of high-frequency return variance than volume alone.